Quiz 3

Graded

Student

Clarisse Cheung

Total Points

3 / 4 pts

Question 1

Question 1 — **2** / 2 pts

1.1 1 — **1** / 1 pt

- ✔ **– 0 pts** Full Credit
- **– 0.1 pts** Majority Credit
- **– 0.5 pts** Half Credit
- **– 1 pt** No Credit
- **– 0 pts** Correct Answer, Minimal Justification
- **– 0 pts** Correct Answer, Incomplete Justification

1.2 2 — **1** / 1 pt

- ✔ **– 0 pts** Full Credit
- **– 0.1 pts** Majority Credit
- **– 0.5 pts** Half Credit
- **– 1 pt** No Credit
- **– 0 pts** Non Shannon Solution

Question 2

Question 2

0 / 1 pt

– **0 pts** Full Credit

– **0.1 pts** Majority Credit

– **0.5 pts** Half Credit

✔ – **1 pt** No Credit

– **0 pts** Incomplete justification

– **0 pts** No explanation on why A is PPT

✔ – **0 pts** A only increases the length of the input by a constant

1 This algorithm is equivalent to $\frac{b(b+1)}{2}$, which only increases the length of the input by a constant

– **0 pts** A does not have sufficient big-O

– **0 pts** A is not PPT

Question 3

Question 3

1 / 1 pt

– **0 pts** Full Credit

– **0 pts** Majority Credit

✔ – **0 pts** Half Credit

– **0 pts** No Credit

2 Then, you can use what you know already about g and h to help you show what is needed.

3 This is a great start! I would proceed by unpacking the definition of $O(\cdot)$ notation and thinking about what needs to be shown to prove that $f \leq O(1/n^c)$.

CMSC 28400 **Quiz 3 (3 questions)** October 13, 2026

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Please write neatly, not too small, and not too faint.

Question 1: In each of the following schemes, $\mathsf{Enc}_k(m) = [m + k \mod 3]$. State in each case whether the scheme is perfectly secret, and justify your answers.

1. $\mathcal{M} = \{0,1\}$ and $\mathcal{K} = \{0,1\}$.

$Enc_k(0) = 0$ with $k=0$ or 1 with $k=1$

$Enc_k(1) = 1$ with $k=0$ or 2 with $k=1$

Take $c = 0$ $\xrightarrow{m \oplus k}$

$Pr[Enc_k(0) = 0] = \frac{1}{2}$

$Pr[Enc_k(1) = 0] = 0$

Since $Pr[Enc_k(0) = 0] \neq Pr[Enc_k(1) = 0]$ then not PI and thus not PS.

2. $\mathcal{M} = \{0,1,2\}$ and $\mathcal{K} = \{0,1\}$.

By Shannon's Thm since $|\mathcal{K}| < |\mathcal{M}|$, this scheme cannot be PS.

↓ states that

$|\mathcal{K}| \geq |\mathcal{M}|$

Question 2: Describe in pseudocode a PPT algorithm $\mathcal{A}$ such that the algorithm $\mathcal{B}$ (below) is not PPT, and justify why $\mathcal{B}$ is not PPT. Let $x \in \{0,1\}^n$ for $n > 0$.

$\mathcal{B}(x)$

$n = |x|$
$z_0 \leftarrow x$
for $i = 1, \ldots, n$ **do**
  $z_i \leftarrow \mathcal{A}(z_{i-1})$
end for
return $(z_1, z_2, \ldots, z_n)$

---

input: size n → $x \in \{0,1\}^n$

loops: n

Algorithm for A:

```
A(b)
 a = 0
 For i = 1, ..., b:
   a += 1
 return a
```

Algorithm A is PPT since dependent on variable b and in worst-case runs a loop b-times. This makes it $O(n)$ time. Since algorithm B has algorithm A within it, it is baseline $O(n)$ time from A. But since B has a loop that runs n times and A runs each time the loop runs, B has $O(n^2)$ time

Question 3: Let $g : \mathbb{N} \to \mathbb{R}_{\geq 0}$ be an arbitrary polynomially-bounded function, and let $h : \mathbb{N} \to \mathbb{R}_{\geq 0}$ be an arbitrary negligible function. Prove that $f(n) = g(n) \cdot h(n)$ is a negligible function.

Def: polynomially bounded if $\exists c \geq 0$ s.t. $f = O(n^c)$

Def: negligibility if $\forall c \geq 0$, $f = O(\frac{1}{n^c})$

$f(n) = g(n) \cdot h(n)$

Need to prove that $\forall c \geq 0$, $g(n) \cdot h(n) = O(\frac{1}{n^c})$.

3

$O(n^c) \cdot O(\frac{1}{n^c})$

Take c

2